



Research paper

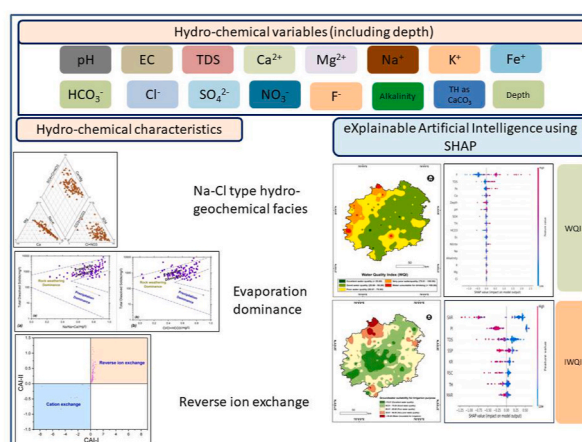
Groundwater hydro-geochemical inferences and eXplainable Artificial Intelligence augmented groundwater quality prediction in arid and semi-arid segment of Rajasthan, India

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HIGHLIGHTS

- Hydro-geochemical inferences of groundwater fused with XAI.
- Reverse ion exchange with positive saturation index indicates Fluoride leaching.
- XGBoost machine learning model able to predicts important features of WQI and IWQI.
- SHAP value interprets the model.
- Fluoride concentration and SAR are most important feature to predict WQI and IWQI.

GRAPHICAL ABSTRACT



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ABSTRACT

Groundwater quality is a crucial aspect especially in the arid and semi-arid segments of the world due to its restricted availability. With increasing consumptions over time period, it is essential to ensure its quality by appraising complex hydro-geochemistry. In the present study, an attempt has been made to evaluate the groundwater hydro-geochemistry in the arid and semi-arid segment of Rajasthan, India and to fill the gap in understanding of groundwater quality by incorporating eXplainable Artificial Intelligence (XAI). 120 groundwater samples were collected during post monsoon season of 2022 and sixteen physico-chemical parameters were analyzed and corresponding inferences were drawn. The hydro-chemical facies indicated Na-Cl composition of groundwater with the dominance of evaporation. Majority of the samples showed reverse ion exchange process along with positive Saturation Index value of Calcite (CaCO₃) and tendency towards leaching F⁻ in the groundwater. Water Quality Index for drinking as well as irrigation purpose showed relatively better quality in the central segment than the marginal region. The SHAP values derived from the XGBoost model depicted

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fluoride (F⁻) as the primary feature influencing overall groundwater quality for drinking purposes, whereas the Sodium Absorption Ratio (SAR) emerged as the key predictor influencing overall groundwater quality for irrigation. The implication of proposed method signifies the importance of incorporating hydro-geochemical inferences with machine learning technique to understand the complex character of groundwater. Further, due to its robustness as well as cost-effectiveness, the application of the method would be helpful in policymaking to safeguard the groundwater resource in arid and semi-arid regions at global scale.

1. Introduction

Accessibility of fresh water for all - has become a global concern as quality and quantity of water is becoming crucial factor with ever increasing population (Adimalla and Li, 2019). The hydro-geochemical characterization of groundwater quality in the spatiotemporal dimension is regarded as a crucial and extensively utilized method by the hydrologists as well as policymakers. (Yawe et al., 2022; Ghosh and Kanchan 2011; Rao et al. 2020, 2021). Further, groundwater evaluation under arid and semi-arid condition grabbed attention of the scholars for

decades as it further underscored the necessity of confirming safe source of water (Andaryani et al., 2022). As identification of groundwater vulnerability is difficult to assess directly, hence number of assessment methods are proposed and can be broadly classified into indexed based methods (Islam and Mostafa, 2022), simulation models (Li et al., 2020), statistical approaches (Swain et al., 2022, Rao et al., 2024) as well as hybrid models (Jamei et al., 2022). The Water Quality Index (WQI) is a popular tool for evaluating water quality because of the advantages, such as the ability to clearly display the overall state of water quality through the index number. WQI is one of the widely used basic

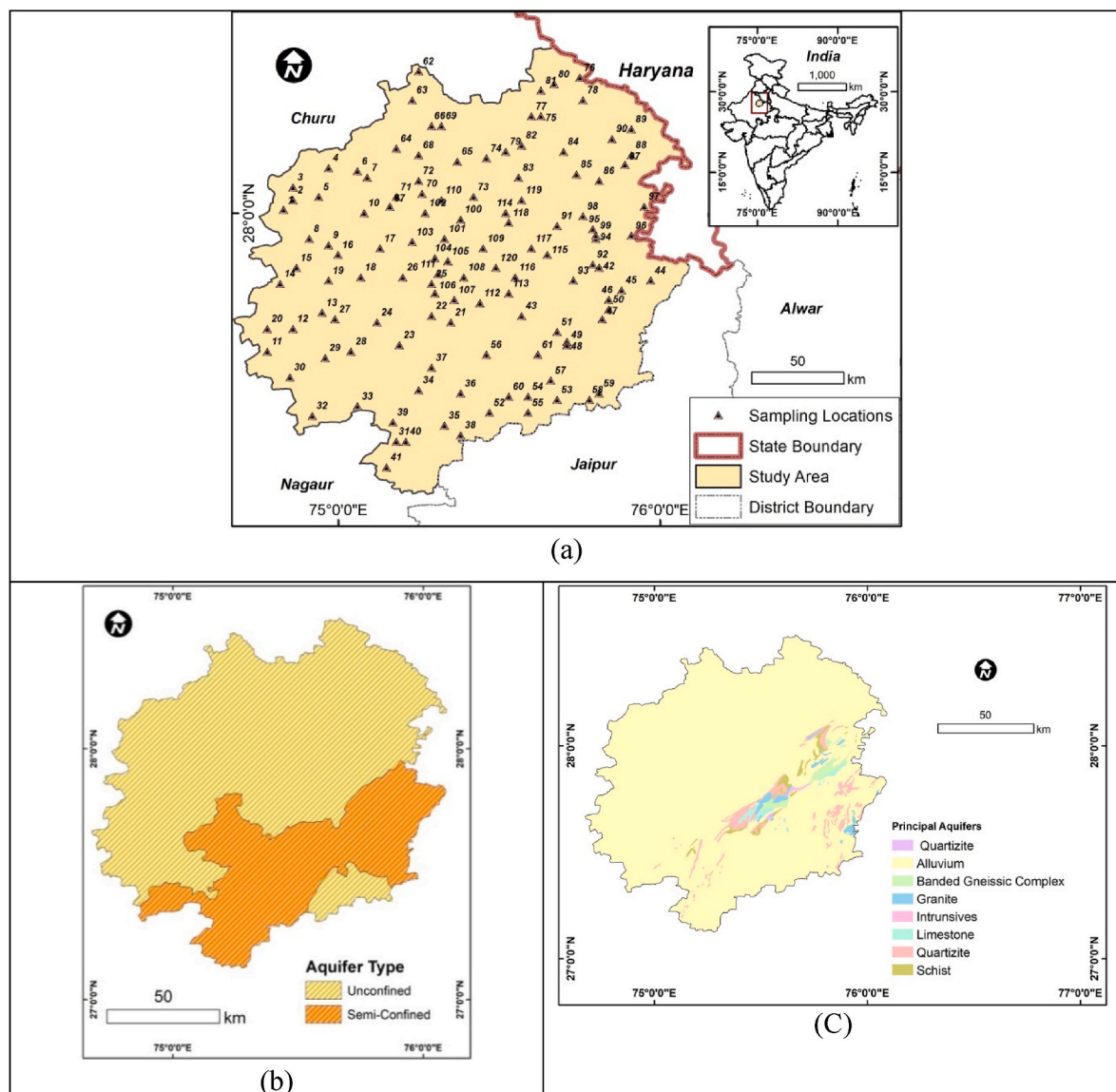


Fig. 1. (a) Location map of the study area, (b) Aquifer map of the study area (Source: [Aquifer Mapping And Ground Water Management, Jhunjhunu District, 2017](#), [District Census Handbook. Village and Town Directory](#); [Aquifer Mapping and Ground Water Management, Sikar 2017](#); [District Census Handbook. Village and Town Directory](#)), (c) Principal aquifers of the study area (Source: [Aquifer Mapping And Ground Water Management, Jhunjhunu District, 2017](#), [District Census Handbook. Village and Town Directory](#); [Aquifer Mapping and Ground Water Management, Sikar 2017](#); [District Census Handbook. Village and Town Directory](#)).

Table 1

1 WQI standards for selected parameters as per BIS.

SR. No.	Water Quality Parameter	BIS Standards(2012)(mg/l)
1	pH	7.5
2	EC	1.5
3	TDS	1000
4	Ca ²⁺	75
5	Mg ²⁺	30
6	Na ⁺	200
7	K ⁺	10
8	Fe	0.3
9	HCO ₃ ⁻	200
10	Cl ⁻	250
11	SO ₄ ²⁻	200
12	NO ₃ ⁻	45
13	F ⁻	1.5
14	Alkalinity	200
15	TH as CaCO ₃	200

BIS (Bureau of Indian Standards) 10500 (2012) Indian Standard Drinking Water Specification.

mathematical method that provides overall water quality situation throughout a region (Selmane et al., 2023; Ravindra et al., 2023). The score of the index indicates the cumulative impact on the overall quality of water. For evaluation of suitability of groundwater for irrigation purposes many studies also focused on the geogenic as well as anthropogenic activities (Gugulothu et al., 2022; Maurya et al., 2023). In the recent years, the focus of the scholars mostly shifted towards supplementing the WQI with Machine Learning (ML) techniques to produce better and less time-consuming models (El Bilali et al., 2021). For the evaluation of model performance, different ML techniques like Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), Multi-Linear Regression (MLR), Deep Neural Network (DNN), Gradient Boosting Machine (GBM) and eXtreme Gradient Boosting (XGBoost) were incorporated (Kouadri et al., 2022). The prediction of the quality reported by the studies showed that ML is associated with the issue of difficulty in explaining the results due to its 'Black box' character (Burrell 2016). To solve the issue of transparency of the functioning of ML, eXplainable Artificial Intelligence (XAI) was introduced (Das and Rad, 2020). By using XAI techniques, more information and deeper understanding about the results for the decision-making is possible. These results are more reliable and transparent, which are essential for making informed decisions in a number of real-world applications (Alshehri and Rahman 2023). SHapley Additive exPlanations or SHAP value is considered to be one of the powerful methods that depict how the machines made prediction of a specific phenomenon (Chakraborty et al., 2021). SHAP is a game-theory-based way to explain the results of any ML model. SHAP explains that the prediction for any sample is equal to the sum of the inputs from its variable values. The variable values work like players in a coalition, and the Shapley value is the average marginal input of a

feature value across all possible coalitions (Moreno-Merino et al., 2022). During rigorous literature survey it was found that presently there are very limited numbers of studies on comprehensive evaluation of groundwater hydro-geochemistry and quality prediction, combined with SHAP based eXplainable Artificial Intelligence (XAI). Hence the present study is an attempt towards filling the critical gap in the understanding of groundwater quality for drinking and irrigation purpose using eXplainable Artificial Intelligence algorithm, augmenting conventional process of groundwater evaluation in the arid and semi-arid region of Rajasthan state, India.

Table 3

Classification of Groundwater quality for irrigation purpose.

	Rating
Excellent	10
Good	8
Poor	6
Very poor	4
Unsuitable	2

Source: Murthy et al. (2010).

Table 4

Descriptive statistics of selected parameters.

Parameters	Minimum	Maximum	Mean	Skewness	Standard Deviation
pH	7.10	8.50	7.78	-0.26	0.32
EC (ds/m)	0.60	11.50	2.06	3.02	1.75
TDS (mg/l)	340.00	8554.00	1396.84	2.87	1219.92
Ca ²⁺ (mg/l)	35.00	260.04	94.86	1.24	41.80
Mg ²⁺ (mg/l)	30.01	223.00	93.62	0.95	33.68
Na ⁺ (mg/l)	25.05	2422.09	366.91	3.22	369.90
K ⁺ (mg/l)	1.51	42.04	4.45	5.86	4.53
Fe (mg/l)	0.00	0.70	0.17	1.42	0.16
HCO ₃ ⁻ (mg/l)	85.41	1122.58	318.93	1.65	169.09
Cl ⁻ (mg/l)	11.38	3545.00	527.78	3.46	530.16
SO ₄ ²⁻ (mg/l)	50.00	1150.00	265.00	1.65	225.25
NO ₃ ⁻ (mg/l)	0.32	483.70	82.80	2.18	96.90
F ⁻ (mg/l)	0.13	4.20	1.14	1.18	0.85
Alkalinity (mg/l)	120.00	1070.00	424.83	1.27	209.11
TH as CaCO ₃ (mg/l)	60.00	2230.00	284.83	4.02	296.90
Depth (mbgl)	11.54	90.00	52.89	0.08	17.63

Source: Computed ds/m-decisiemens per metre mg/l-milligram per litre, mbgl-meters below ground level

Table 2

Selected indices for Irrigation Water Quality Index and respective description.

Indices	Acronym	Formula	Reference	Formula No.
Sodium adsorption ratio	SAR	$SAR = \frac{Na^+}{\sqrt{\frac{Ca^{2+} + Mg^{2+}}{2}}}$	Richards, 1954	(15)
Magnesium Absorption Ratio	MAR	$MAR = \frac{Mg^{2+}}{Ca^{2+} + Mg^{2+}} \times 100$	Paliwal (1972)	(16)
Residual Sodium Carbonate	RSC	$RSC = (CO_3^{2-} + HCO_3^-) - (Ca^{2+} + Mg^{2+})$	Eaton (1950)	(17)
Kelly's Ratio	KR	$KR = \frac{Na^+}{Ca^{2+} + Mg^{2+}}$	Kelly (1954)	(18)
Permeability Index	PI	$PI = \frac{((Na^+ + K^+) + \sqrt{HCO_3^-})}{(Na^+ + Ca^{2+} + Mg^{2+})} \times 100$	Doneen (1962)	(19)
Soluble Sodium Percent	SSP	$SSP = \frac{Na^+}{Ca^{2+} + Mg^{2+} + Na^+ + K^+} \times 100$	Richards, 1954	(20)
Total Hardness as CaCO ₃	TH	TH = 2.5Ca + 4.1 Mg	Todd (1980)	(21)
Total Dissolved Solids	TDS	Directly calculated using TDS meter		

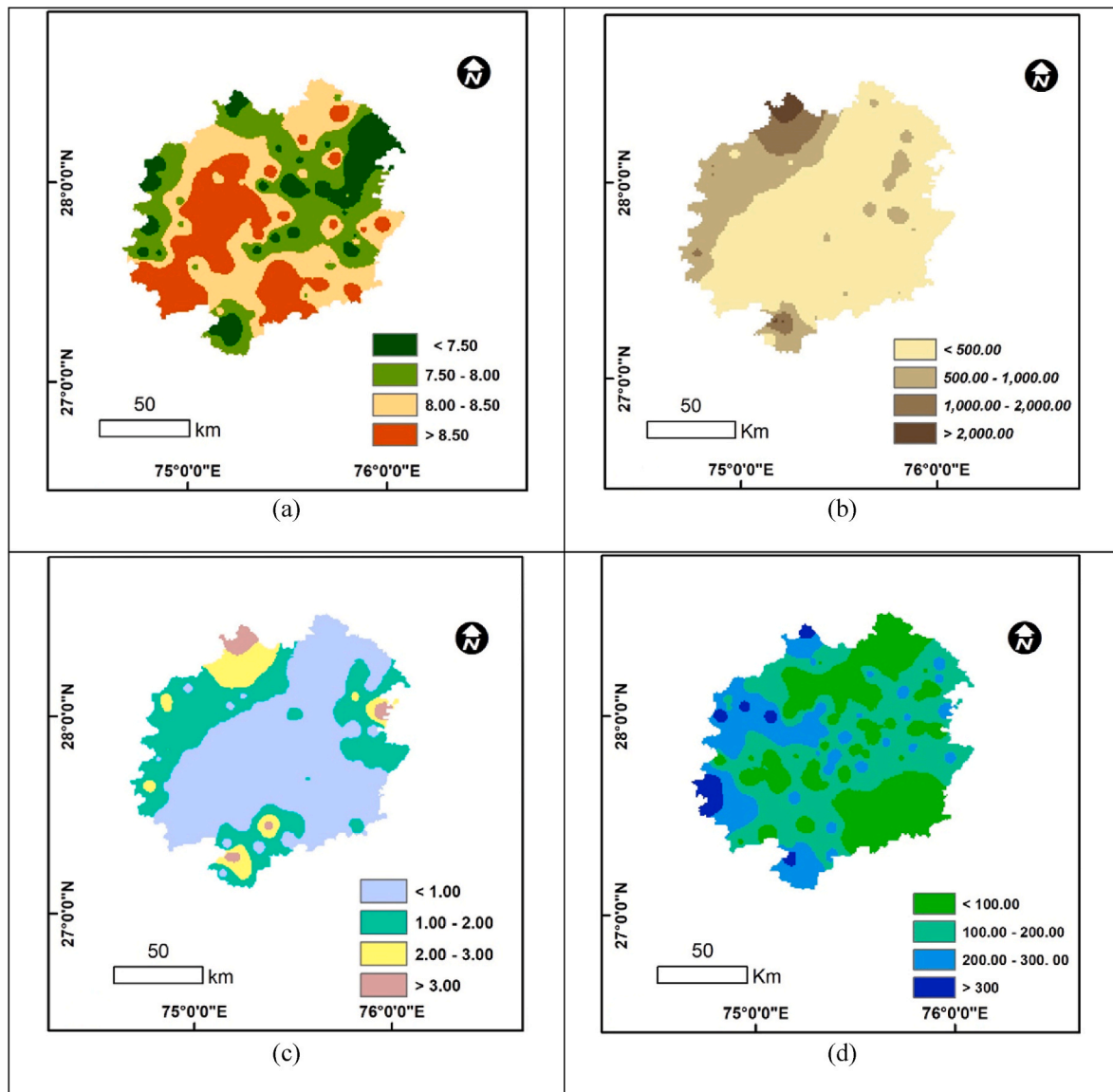


Fig. 2. Spatial distribution of physicochemical parameters (a) pH, (b) TDS, (c) EC, (d) Ca²⁺, (e) Mg²⁺, (f) Na⁺, (g) K⁺, (h) Fe, (i) HCO₃⁻, (j) Cl⁻, (k) SO₄²⁻, (l) NO₃⁻, (m) F⁻, (n) Alkalinity, (o) TH as CaCO₃, (p) Depth. (The unit of EC is in ds/m and unit of Depth in in mbgl. pH is unitless and rest of the parameters are in mg/l).

2. Materials and methods

2.1. Study area description

The region under investigation is a portion of the Transitional Plain of inland drainage (ACZ-III) agro climatic zone in the western part of the Indian state of Rajasthan (Kar, 2014). Geographically the region is located between 27° 21'N and 28° 31'N in latitude and 74° 44'E and 76° 06'E in longitude and spans 13,660 km² area (Fig. 1a). It shares the state border with Haryana in the north-east and surrounded by the districts of Rajasthan in the north-west with Churu, in south-west by Nagaur, in south by the Jaipur and in East by the Alwar districts. From the physical point of view, the Aravalli range runs through the south eastern and southern segment of study area with an elevation range of 292m–1026m (Saxena, 2019). In the north, the area is almost flat, with rolling hills that range in height from 300m to 450 m (District Census Handbook. Village and Town Directory of Sikar, 2011a; District Census Handbook. Village and Town Directory of Jhunjhunu, 2011b). Climatically the entire area is known for hot summers and cold winters with little rain.

The northern segment is mostly associated with arid type of climate while southern portion showed semi-arid type of climate (District Census Handbook. Village and Town Directory; Hydrological Atlas of Rajasthan, 2013a, Hydrological Atlas of Rajasthan, 2013b).

2.2. Hydrogeological setup

The study area comprises two major aquifer systems. Most of the northern and a small patch in southern portion of the study area is associated with unconfined aquifer while the central portion of the study area is related to semi-confined aquifer (Fig. 1b). A thick layer of alluvium, deposited on top of bedrock, and both alluvium and hard rocks can form groundwater. Aquifers formed in quartzite rocks because of the joints and cracks. The younger alluvium makes up only 2% of the aquifer area, but the older alluvium is the most common, with about 76% of the area spread across the entire western half as well as in between quartzite aquifers in the eastern part (Fig. 1c). The aquifer condition of the northern segment of the study region formed with quartzite and alluvium as the primary aquifer material. Here, older alluvium accounting

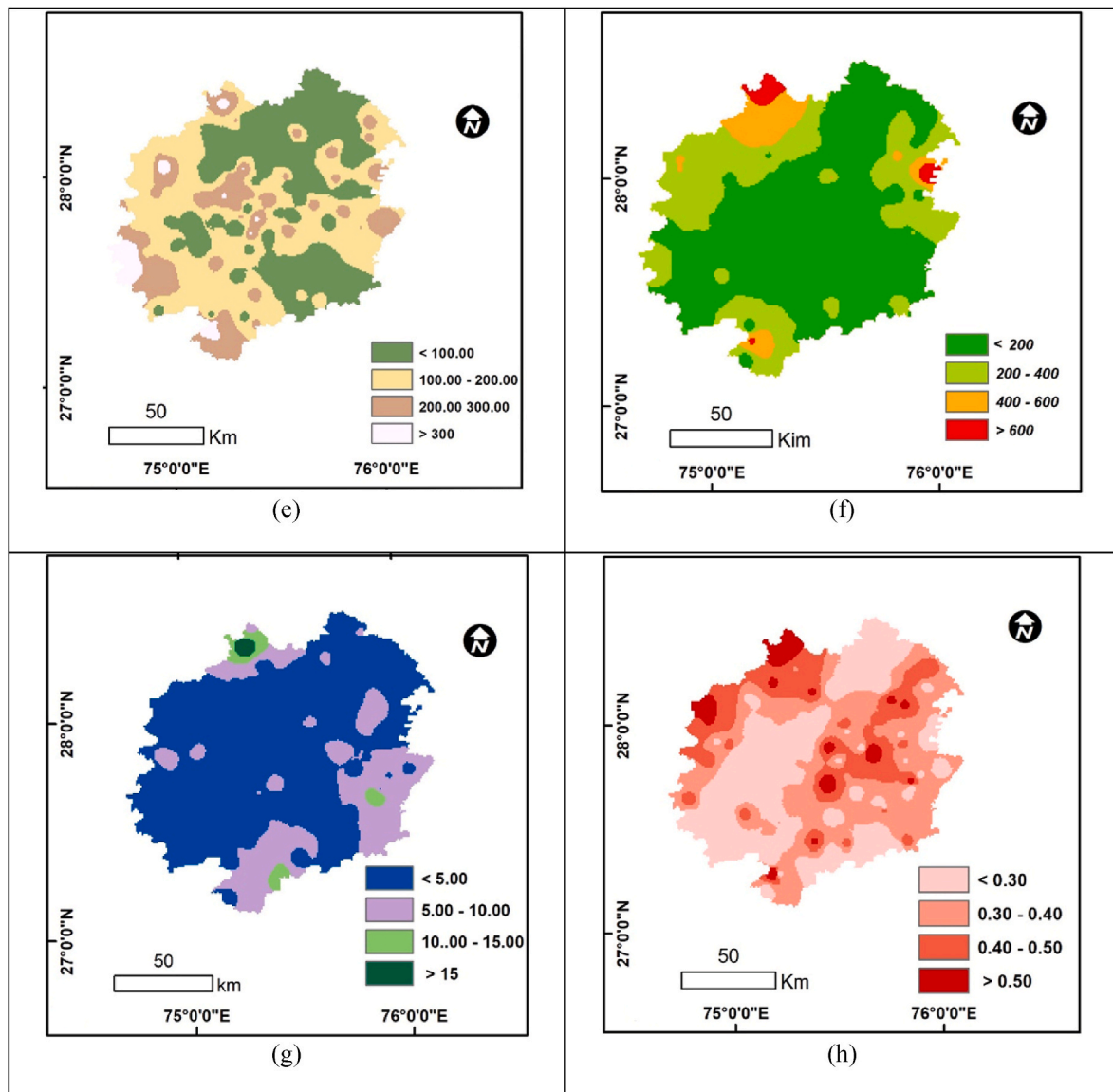


Fig. 2. (continued).

for 76% of the aquifer area and younger alluvium accounting for the remaining 6%. Approximately 14% of aquifers are comprised of fractured and weathered quartzite adjacent to slopes in the south-eastern segment. (Hydrological Atlas of Rajasthan, 2013a, Hydrological Atlas of Rajasthan, 2013b). Water level in the study area during pre-monsoon season showed as low as 40 mbgl while transmissivity varied between 100 m²/day to 1915 m²/day in the north and 495 m²/day (mean) in the south. Water yield in the north varies from 252 m³/day to 1296 m³/day while in the south, it ranged from 10 m³/day to 30 m³/day. Moderate water yield was noticed in areas where the weathered mantle is covered by a substantial layer of blown sand (Groundwater Brochure, 2008, Jhunjhunu District Census Handbook. Village and Town Directory; Groundwater Scenario, 2013, Sikar District Census Handbook. Village and Town Directory; Part XII-A & B, Series-9, 2011b; Ground Water Year Book, 2021–2021).

2.3. Sample collection

120 groundwater samples from dug wells, tube wells and shallow well pumps were collected during post monsoon season of 2022. Selection of sampling locations was based on random sampling technique

and maximum uniform spatial coverage (Fig. 1a). The tube-wells and shallow well pumps were flushed for five to 10 min to remove any stagnant water before collection of the water samples. Samples were collected in pre-cleaned 500 ml polyethylene bottles. Each sample container was labelled with water-resistant marker and information regarding each sample was recorded. Google maps mobile GPS was used to mark the sampling locations with unique IDs. For the fundamental understanding about the physico-chemical characteristics of groundwater, sixteen parameters viz. pH, Electrical Conductivity (EC)(ds/m), Total Dissolved Solids (TDS)(mg/l), Calcium (Ca²⁺) (mg/l), Magnesium (Mg²⁺) (mg/l), Sodium (Na⁺) (mg/l), Potassium (K⁺) (mg/l), Iron (Fe) (mg/l), Bicarbonate (HCO₃⁻) (mg/l), Chloride (Cl⁻) (mg/l), Sulfate (SO₄²⁻) (mg/l), Nitrate(NO₃⁻) (mg/l), Fluoride (F⁻) (mg/l), Alkalinity (mg/l), Total Hardness as CaCO₃ (would be mentioned as TH in the rest of the paper)(mg/l) and Depth (mbgl) were selected.

2.4. Chemicals and laboratory work

Double distilled water and analytical grade chemicals were used in the entire process in analysing Na⁺, Ca²⁺, Mg²⁺ and K⁺ through standard Flame photometric technique (Rice et al., 2012) (Systronics flame

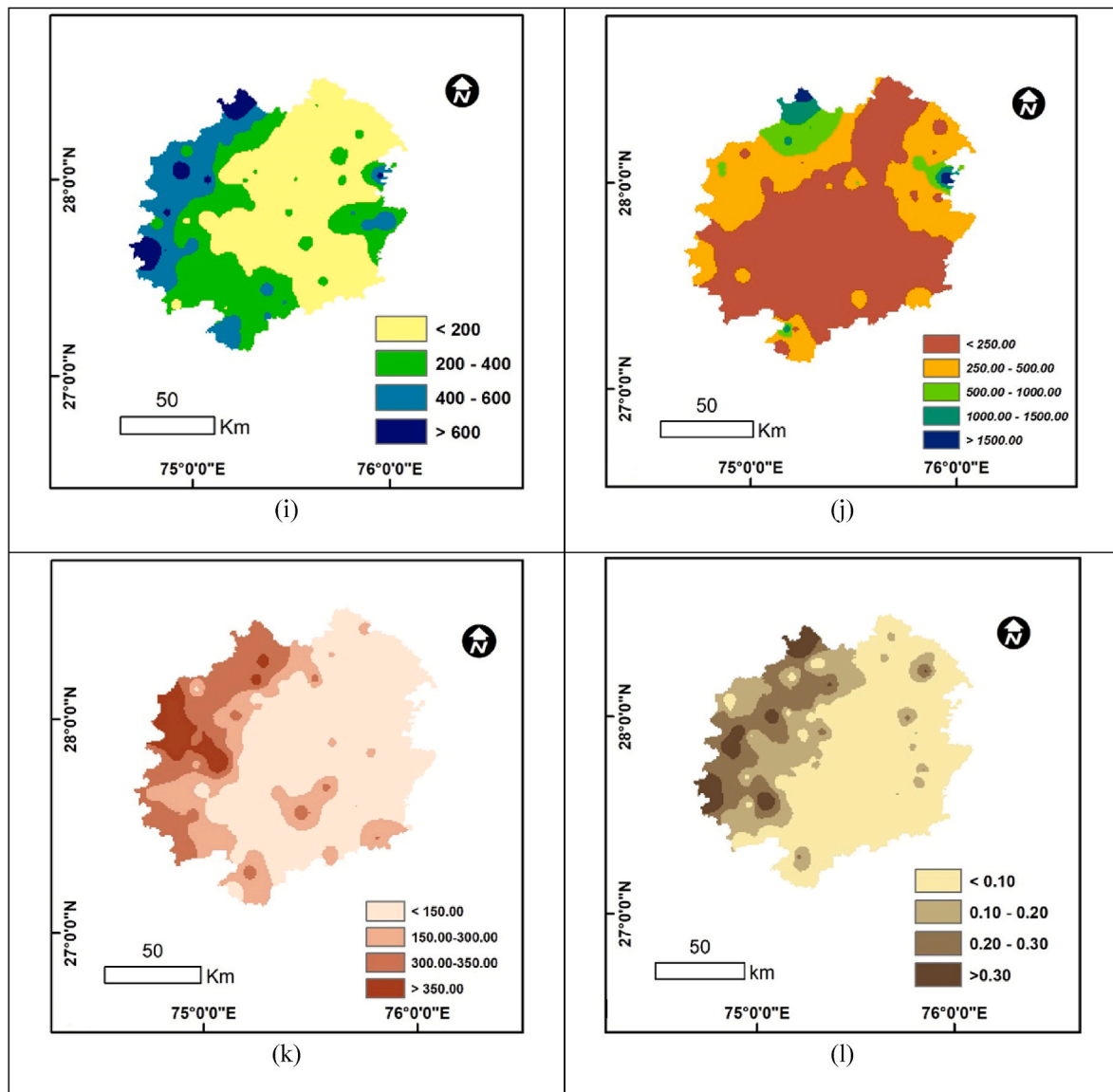


Fig. 2. (continued).

photometer, Model 128, Compressor 126). For the preparation of standard solution Ca (Himedia, India), Calcium Carbonate (CaCO_3) salt was used. For the determination of Na^+ and K^+ , similar technique was used using Potassium chloride (KCl) and Sodium Chloride (NaCl) salts (Himedia, India) for the preparation of standard solutions.

2.5. Analysis of physico-chemical parameters

Parameters like pH, EC, TDS, SO_4^{2-} and Fe were tested on-field using digital pH meter (Hanna make, model HI 9827), EC meter (Hanna make, DiST 4, Model no HI 98303) and TDS meter (Hanna make, DiST 1, Model HI 98300). Fe and SO_4^{2-} were tested using AQUASOL field water testing kits (Aquasol Iron Test kit, AE-303, Range: 0–2 mg/l and Aquasol Sulfate Test kit, AE-209, Range: 10–200 & 50–1000 mg/l). The collected samples were immediately transported to the laboratory for the analysis of rest of the parameter. Parameters like F^- , NO_3^- , TH, and alkalinity were tested in Public Health Engineering Department (PHED) laboratory, Sikar and Jhunjhunu, Rajasthan, India, while Parameters like Mg^{2+} , HCO_3^- and Cl^- were analyzed in the State Level Water Testing Laboratory, Jaipur, Rajasthan, India. Na^+ , Ca^{+} , and K^+ were tested in Research laboratory, School of Liberal Arts and Sciences, Mody University of

Science and Technology, Lakshmangarh, Sikar, Rajasthan using standard flame photometric technique (Rice et al., 2012).

2.6. Water quality index (WQI) for drinking purpose

The Water Quality Index (WQI) was developed by (Horton, 1965) and then used by Brown et al. (1970). In the present work the WQI method developed by Horton (1965) was taken as it is. It is a useful technique for evaluating the overall quality of groundwater (Acharya et al., 2018; Adimalla and Qian 2019), which turns a lot of information about the quality of groundwater into a single value and makes it easier to understand the information. In this work, WQI was calculated using this way (1–4):

$$Wi = \frac{Wi}{\sum_{i=1}^n wi} \quad (1)$$

Where Wi : represents the relative weight, wi : represents the weight of each parameter, while n represents the number of parameters.

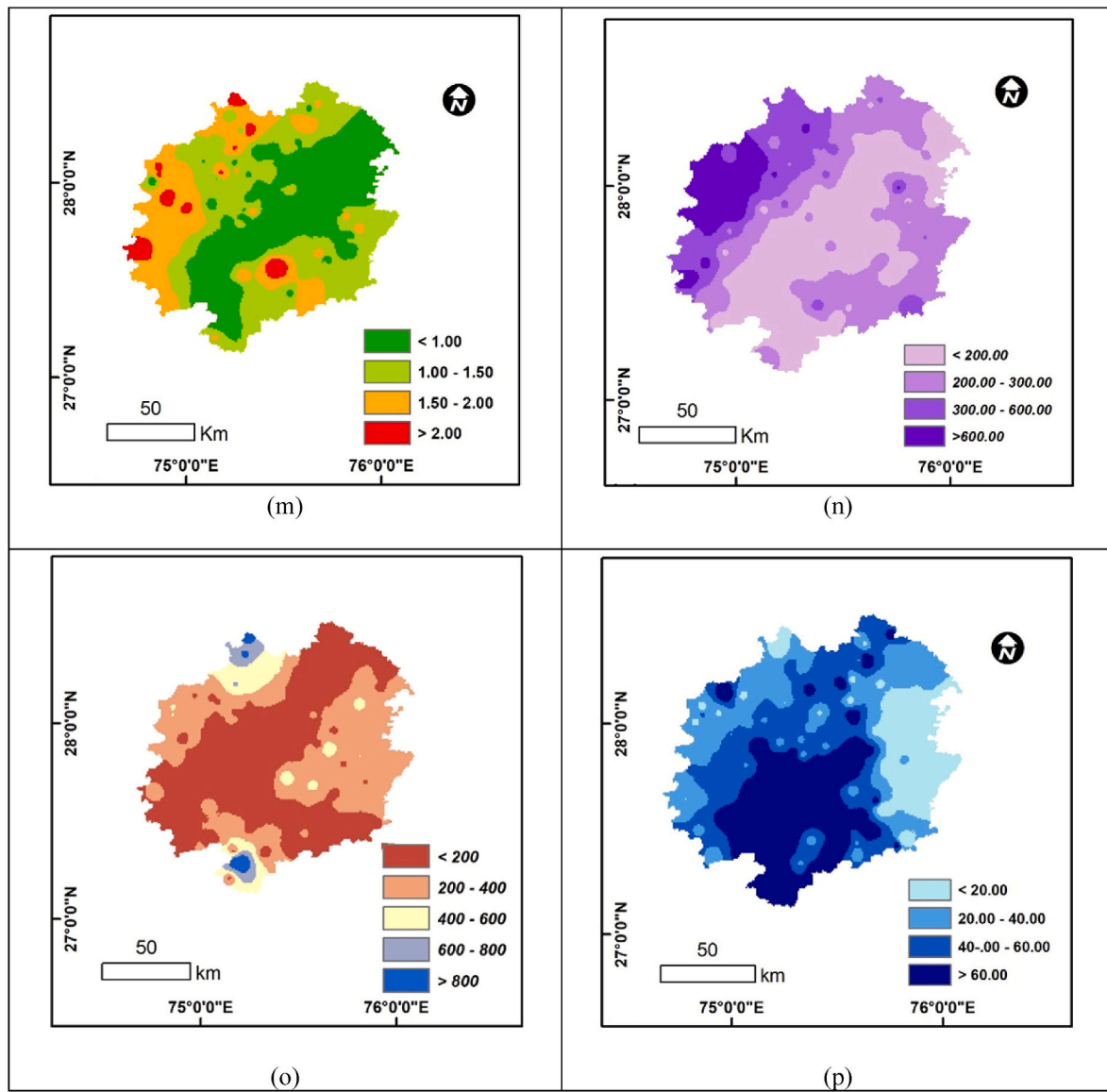


Fig. 2. (continued).

$$Q_i = \frac{C_1 - C_{ip}}{S_i - C_{ip}} \times 100 \quad (2)$$

Q_i : is the rating based on the "ith" parameter's concentration, C_1 : is the chemical parameter concentration in each groundwater sample, C_{ip} : is the ideal parameter value in purified water (consider $C_{ip} = 0$ for all, except pH where C_{ip} of pH = 7) S_i : represents the standard value (BIS IS 10500, 2012) (Table 1) for each chemical parameter.

$$SI_i = W_i \times Q_i \quad (3)$$

SI_i : represents the sub-index of the "ith" parameter.

$$WQI = \sum_{i=1}^n SI_i \quad (4)$$

2.7. Irrigation water quality index (IWQI)

In the present work, the standard given by Murthy et al. (2010) was taken into considerations for evaluation of Irrigation Water Quality Index (IWQI). Eight crucial indices related to irrigation water quality were incorporated (Table 2). For integrating all of the indices, rates were

assigned to each of the class of the individual indices (Murthy et al., 2010). As all the selected indices were equally significant in controlling the overall water quality for irrigation purpose, hence, all the indices were assigned with similar rates. Table 3 showed the ratings of different classes of water quality for irrigation purpose. Once all the indices were reclassified according to the ratings, the following formula (5) was used for depicting final irrigation water quality index (IWQI):

$$IWQI_x = \sum_{i=1}^8 (R_x) \quad (5)$$

Here $IWQI_x$ is irrigation water quality index,

R_x = Rating factor for the parameters.

The result obtained from the WQI and IWQI analysis were incorporated in the GIS environment for illustrating its spatial distribution.

2.8. eXplainable artificial intelligence

As an application of Artificial Intelligence (AI), Machine Learning (ML) is becoming more dependable over the time period (Sajedi-Hosseini et al., 2018; Agrawal et al., 2021). Automation, efficient data

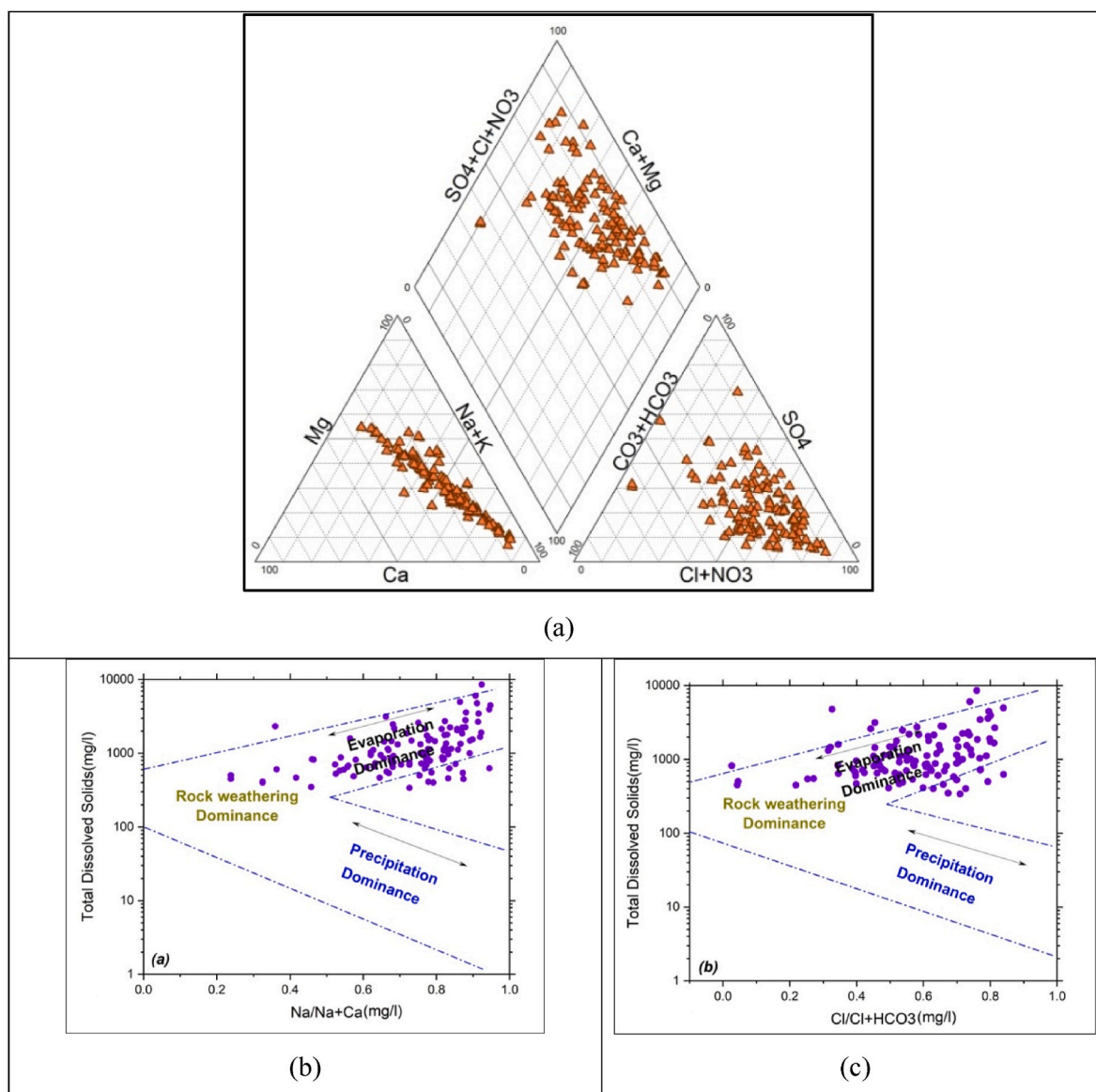


Fig. 3. (a) Piper diagram showing distribution of water chemistry based on major ions, (b) Cations ratio and (c) Anions ratio of groundwater using Gibbs plot.

Table 5

Correlation matrix of selected physico-chemical parameters.

	pH	EC	TDS	Ca ²⁺	Mg ²⁺	Na ⁺	K ⁺	Fe	HCO ₃ ⁻	Cl ⁻	SO ₄ ²⁻	NO ₃ ⁻	F ⁻	Alkalinity	TH	Depth
pH	1															
EC	-0.28	1														
TDS	-0.29	0.80	1													
Ca ²⁺	-0.08	0.42	0.40	1												
Mg ²⁺	-0.01	0.30	0.23	0.83	1											
Na ⁺	-0.29	0.93	0.79	0.34	0.21	1										
K ⁺	-0.22	0.33	0.39	0.13	0.18	0.31	1									
Fe	-0.37	0.40	0.52	0.27	0.20	0.38	0.29	1								
HCO ₃ ⁻	-0.05	0.68	0.63	0.65	0.49	0.62	0.24	0.21	1							
Cl ⁻	-0.26	0.88	0.75	0.35	0.22	0.94	0.31	0.43	0.56	1						
SO ₄ ²⁻	-0.05	0.36	0.49	0.21	0.08	0.35	0.09	0.15	0.41	0.33	1					
NO ₃ ⁻	-0.04	0.35	0.49	0.41	0.21	0.34	0.27	0.20	0.52	0.30	0.32	1				
F ⁻	0.16	0.26	0.35	0.11	-0.01	0.23	-0.02	-0.08	0.44	0.24	0.46	0.22	1			
Alkalinity	0.03	0.31	0.45	0.29	0.15	0.28	0.03	0.14	0.50	0.27	0.57	0.46	0.51	1		
TH as CaCO ₃	-0.46	0.68	0.80	0.30	0.25	0.66	0.42	0.54	0.37	0.54	0.26	0.27	-0.03	0.07	1	
Depth	0.18	-0.26	-0.20	-0.10	-0.06	-0.24	-0.14	-0.13	-0.22	-0.28	-0.13	-0.06	-0.10	-0.25	-0.07	1

Source: Computed

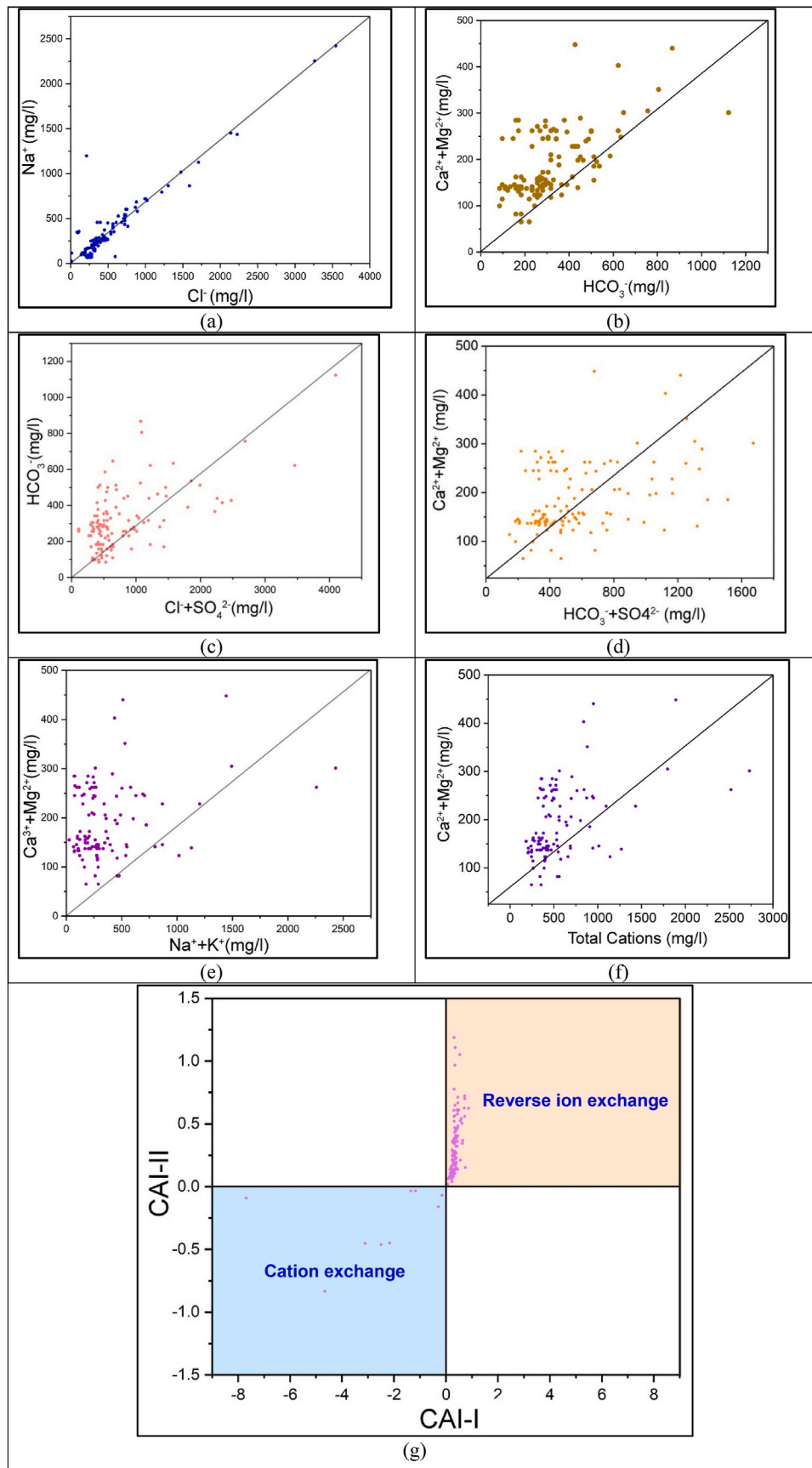


Fig. 4. Ionic relation between (a) Na^+ vs Cl^- , (b) $(\text{Ca}^{2+} + \text{Mg}^{2+})$ vs HCO_3^- , (c) HCO_3^- vs $(\text{Cl}^- + \text{SO}_4^{2-})$, (d) $(\text{Ca}^{2+} + \text{Mg}^{2+})$ vs $(\text{HCO}_3^- + \text{SO}_4^{2-})$, (e) $(\text{Ca}^{2+} + \text{Mg}^{2+})$ vs $(\text{Na}^+ + \text{K}^+)$, (f) $(\text{Ca}^{2+} + \text{Mg}^{2+})$ vs Total cations, (g) Schoeller's ion exchange reactions.

Table 6
Chloro-alkaline index for CAI-1 and CAI2.

	Minimum	Maximum	Mean
Chloro-alkaline index			
CAI-1	-7.68	0.87	0.13
CAI-2	-0.83	1.19	0.28

Source: Computed

Table 7
Saturation Index(SI).

	Minimum	Maximum	Mean
Halite	-5.49	-0.92	-2.98725
Gypsum	0.05	383	3.9275
Calcite	1.98	3.86	2.987

Source: Computed

handling and wide range of applications are some of the key features for its popularity in all dimensions of science and technology (Ashima et al., 2021). On one hand its accuracy level is significant in predicting certain phenomena at the same time it suffers from its complex 'Black box' nature. eXplainable Artificial Intelligence or XAI solve the issue by incorporating both accuracy as well as fidelity so that the model can explain the underlying process of 'Black box' (Alshehri and Rahman, 2023).

2.9. XGBoost and SHAP

XGBoost has emerged as a highly prevalent and effective ML technique in contemporary times (Chen and Guestrin, 2016). XGBoost is a popular gradient boosting approach for machine learning tasks, such as classification and regression issues. The approach employs a sequence of tree-based regressors that demonstrates superior predictive capabilities (Jing et al., 2023). Understanding the contributions of each feature to the predictions provided by XGBoost is made possible using SHAP values (Xia et al., 2022). The SHAP values assess the contribution of each characteristic to the prediction and a well-known mathematical term in game theory (Shapley 1953, Bedi et al., 2020). The decision-makers' ability to comprehend the linkages between inputs and outputs is facilitated by the interpretability of the SHAP values and their clear description of the decision-making process (Moreno-Merino et al., 2022).

Consider the dataset used to train the regression model is $W = [p_i, q_i; i = 1 \text{ to } n]$, here p_i is independent variable and q_i is a dependent variable (actual target value) (Expression (6)):

$$q'_i = \sum_{k=1}^k v_k(p_i), v_k \in F \quad (6)$$

Here, v_k is k th regression tree and $v_k(p_i)$ is the output of v_k for i th input sample from dataset W and q'_i is the predicted target value. To train the regression tree v_k the following objective function is used (Expression 7 and 8)-

$$Obj = \sum_{i=1}^n m(q_i, q'_i) + \sum_{k=1}^n n(v_k) \quad (7)$$

$$n(v_k) = \delta T + \frac{1}{2} \varepsilon [[\omega]]^2 \quad (8)$$

Here, function 'm' computes the loss using actual, predicted value. ε and T are regularization arguments to evade overfitting of the regressor model (Patel et al., 2022).

In SHAP, the importance of feature (k) for model outcome $f; (f)$, is assigned weight for feature contribution summary towards output of model (x_i) for the overall possible combinations of features. The

expression for (f) can be expressed as the following formula (9):

$$\phi^k(f) = \sum_{S \subseteq \{x^1, \dots, x^p\} / \{x^k\}} \frac{|S|! (p - |S| - 1)!}{p!} (f(S \cup \{x^k\}) - f(S)) \quad (9)$$

Here, S = features subset, p = feature number in model, and x_j = feature j .

2.10. Interpolation, mapping and machine learning

In the present study, OriginPro 2022b was used for the preparation statistical diagrams. ArcGIS software was used for the production of thematic maps. The base map of the study area was prepared using District wise hydrological atlas of Sikar and Jhunjhunu (Hydrological Atlas of Rajasthan, Jhunjhunu District Census Handbook. Village and Town Directory; Hydrological Atlas of District Census Handbook. Village and Town Directory; Part XII-A & B, Series-9, 2011a). Selected physico-chemical parameters and irrigation indices of all the sample locations were incorporated in the GIS environment along with their corresponding normalized values. For the generation of spatial distribution maps, Inverse Distance Weightage (IDW) method was applied. For the preparation of the WQI map, the results were incorporated for the respective samples followed by interpolation of maps using IDW with classes viz. excellent water quality, good water quality, poor water quality, very poor water quality and water unsuitable for drinking purpose. For the preparation of the IWQI map, spatial distribution maps of indices of irrigation were reclassified with their corresponding ratings. The task was performed through *reclassification* tool in ArcGIS. Then each of the reclassified layers were converted into polygon and followed by *intersection* of all the layers. Then in the final layer, rating column of each of the indices were summed up through *geometry calculator* to generate final ratings and classified with five classes. Python programming language was used on the notebook of Google Colab to run the XAI model and respective graph generation. The module used in the coding for the depiction of order of importance of feature in terms of model performance was *xgboost* with the function of *xgb.XGBRFRegressor*. PHREEQC software package (Parkhurst and Appelo, 1999) was used to depict the SI values that helped in giving insights into the geochemical processes.

3. Results and discussion

3.1. Spatial distribution of physicochemical parameters

Descriptive statistics of selected physico-chemical parameters are shown in Table 4. The pH was rather alkaline throughout the study region. The central and southern segments showed higher pH levels, while north eastern and western portions showed pH levels below 7.50 (Fig. 2a). Western, north-western, and southern segments of the research area exhibited higher TDS levels than the central part (Fig. 2b). Central segment showed lower level of EC than the western, north-western, eastern, and southern segments (Fig. 2c). Concentration of Ca^{2+} and Mg^{2+} were similar. The central region showed higher concentration of Ca^{2+} and Mg^{2+} than the northern and eastern portions (Fig. 2d and e). The centre portion showed lower Na^+ concentration than the north-eastern and north-western parts (Fig. 2f). Smaller patches in the north west and north east showed higher concentration of K^+ (Fig. 2g). The result depicted the fact that groundwater cations in decreasing the following order: $Na^+ > Ca^{2+} > Mg^{2+} > K^+$. North-western, western, and eastern regions showed higher level of Fe concentrations, whereas the central region showed the low level (Fig. 2h). Western portion of the study area showed the high level of HCO_3^- concentration, whereas the middle portion showed lower level (Fig. 2i). Western and north-eastern regions showed highest concentration of Cl^- than the central regions (Fig. 2j). Entire west segment showed higher concentration of SO_4^{2-} while the middle and southern parts had relatively lesser concentration

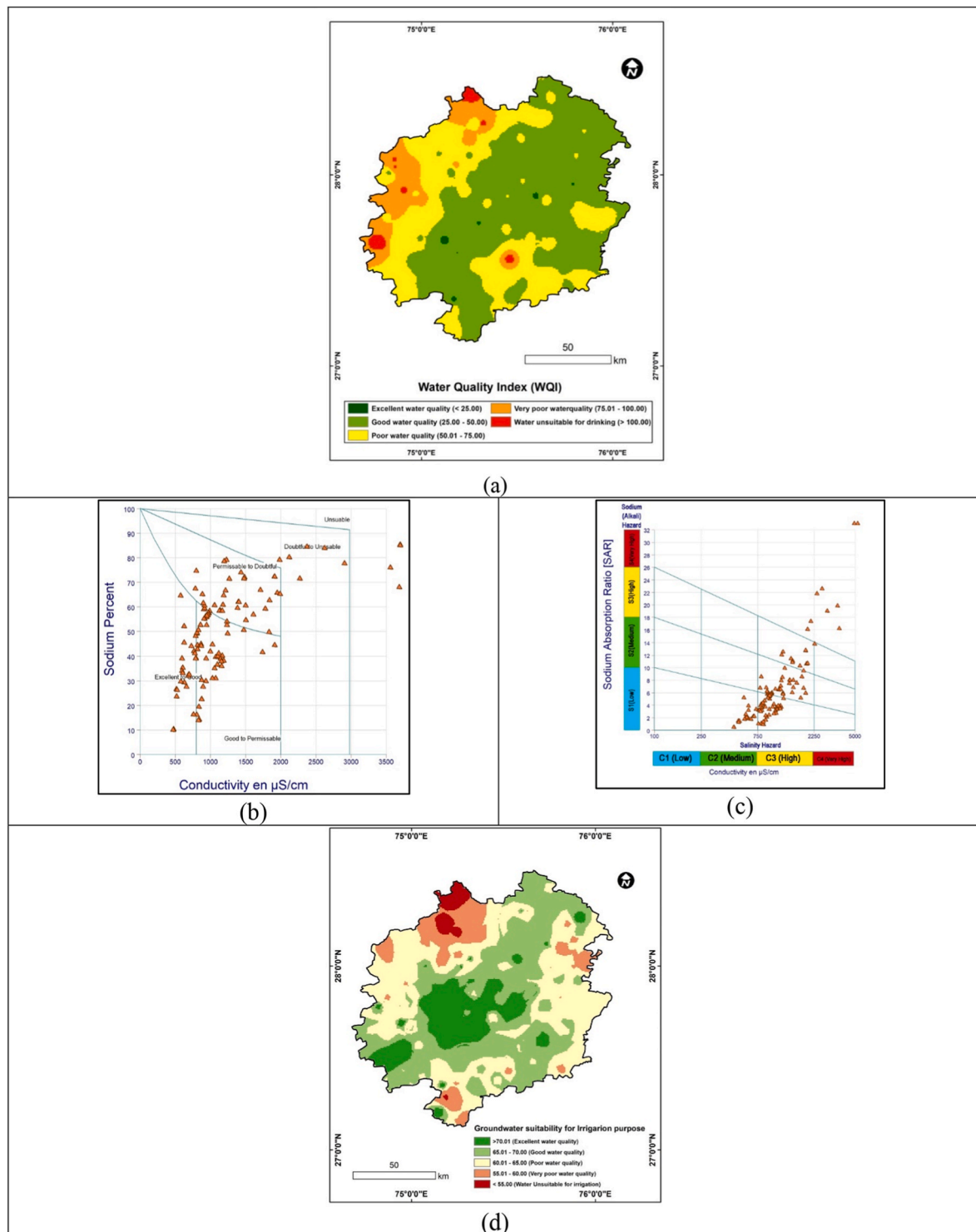


Fig. 5. (a) Groundwater Quality categorization for drinking purpose (WQI), (b) Sodium percent (Na%) and (c) a Sodium adsorption ratio (SAR) using Wilcox diagram (d) Groundwater Quality categorization for irrigation purpose (IWQI).

(Fig. 2k). Western segment showed higher concentration of NO_3^- than the middle and western segments (Fig. 2l). F^- concentration was highest in the eastern and western periphery regions and lowest in the middle area (Fig. 2m). Western part of the research region showed higher alkalinity than the middle part (Fig. 2n). Northern and southern edge had higher concentration of TH than the central and southern part (Fig. 2o). Eastern and western periphery depicted lesser water depth than the middle and southern segment (Fig. 2p). Groundwater anionic

concentration in decreasing order can be depicted in the following manner: $\text{Cl}^- > \text{HCO}_3^- > \text{SO}_4^{2-} > \text{NO}_3^- > \text{F}^-$. Elevated Cl^- and HCO_3^- levels might cause into groundwater mineral dissolution (Marghade et al., 2021; Piper 1944).

3.2. Hydro-geochemical facies

Piper diagram is generally used to study the hydro-geochemistry of

Table 8

Descriptive statistics of indices for irrigation.

	Mean	Max	Min	SD	Skewness
SAR	13.45	69.80	1.01	11.79	2.49
RSC	130.46	821.56	-147.36	137.45	1.30
MAR	50.28	66.37	33.56	5.70	0.49
KR	2.05	8.60	0.16	1.72	1.90
PI	62.71	90.86	22.78	16.65	-0.49
SSP	58.39	89.45	13.70	17.58	-0.45
TDS	1396.84	8554.00	340.00	1219.92	2.87
TH	284.83	2230.00	60.00	296.90	4.02

Source: Computed

water that is controlled by anions and cations (Ma et al., 2022). Central diamond segment depicted that majority of the samples are in the NaCl type as well as mixed CaMgCl type. Only a few samples are found in the CaCl type of segment (Fig. 3a). In the study area significant number of samples exhibited Na-Cl type. A strong positive correlation between Na^+ and Cl^- ($r = +0.94$) (Table 5) also indicated of strong association between both. On the other hand, all the selected parameters showed relatively weak correlation with depth in specific (Table 5). The water facies in the present study area indicated an analogous kind of water,

went through analogous geogenic host rock and water interaction.

3.3. Water-rock interaction

Gibbs (1970) diagram can offer significant inferences on the relationship between chemical composition of water and the lithological features of an aquifer (Sunkari et al., 2023). The relation between TDS and $\text{Na}/(\text{Na} + \text{Ca})$ revealed that, majority of samples were in the upper portion of the graph, showing evaporation as the dominant factor affecting groundwater chemistry (Fig. 3b). On the other hand, relation between TDS and $\text{Cl}^-/(\text{Cl} + \text{HCO}_3)$ showed rock weathering and evaporation as dominant factors affecting groundwater chemistry (Fig. 3c). Most groundwater samples were into the evaporation dominance zone, with only a few samples were into the rock weathering dominance zone. The climatic condition of the present study area confirms the depicted result and substantiated by other previous studies with similar climatic and geogenic condition (Loni et al., 2015).

3.4. Impact of weathering of rock on groundwater chemistry

Two important mechanisms related to impact of weathering of rock on groundwater chemistry in the present context are cation exchange

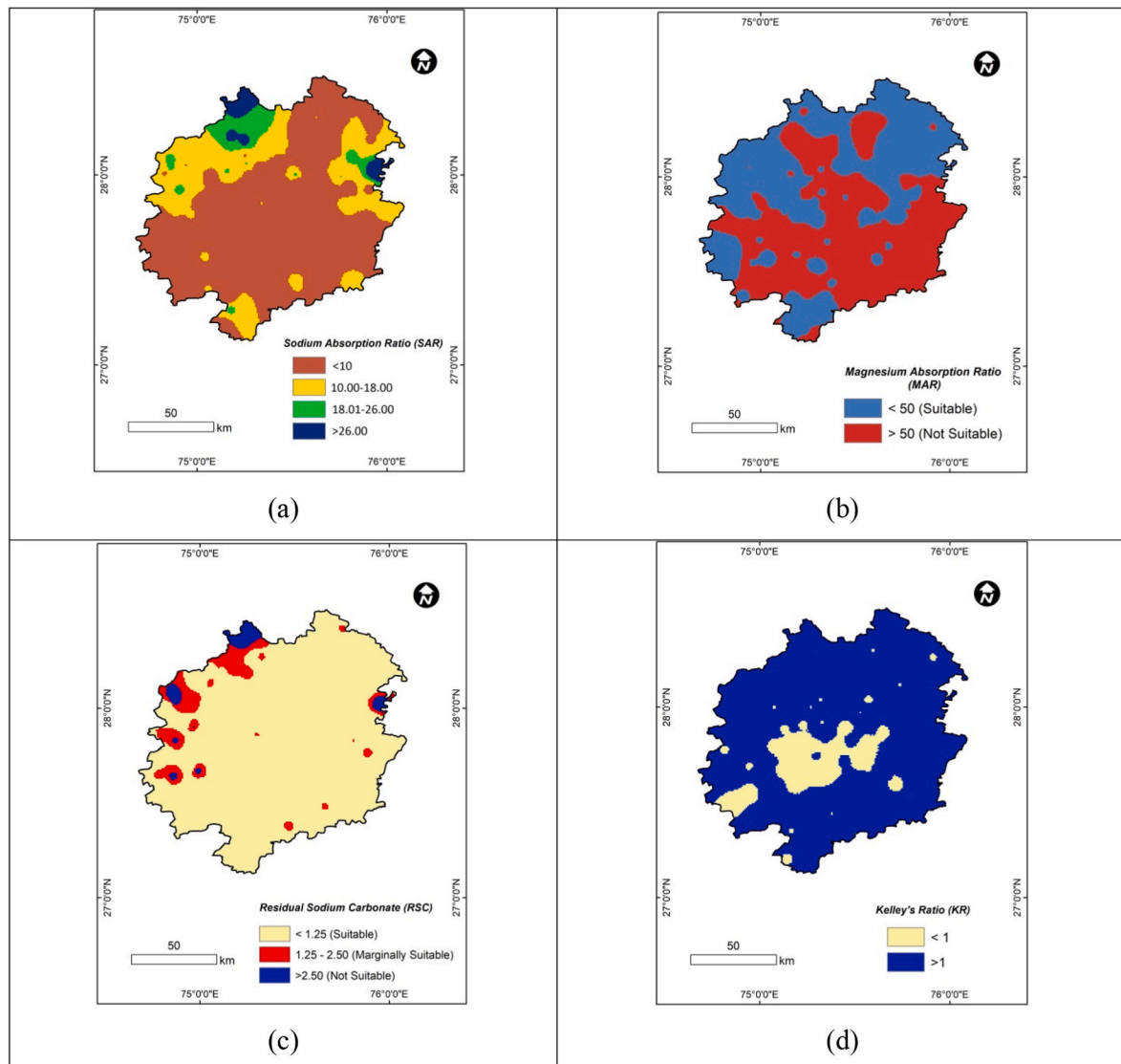


Fig. 6. Spatial distribution of (a) Sodium Absorption Ratio (SAR), (b) Magnesium Absorption Ratio (MAR), (c) Residual Sodium Carbonate (RSC), (d) Kelly's Ratio (KR), (e) and (f) Permeability Index (PI), (g) Soluble Sodium Percent (SSP), (h) Total Hardness as CaCO_3 , (i) Total Dissolved Solids (TDS).

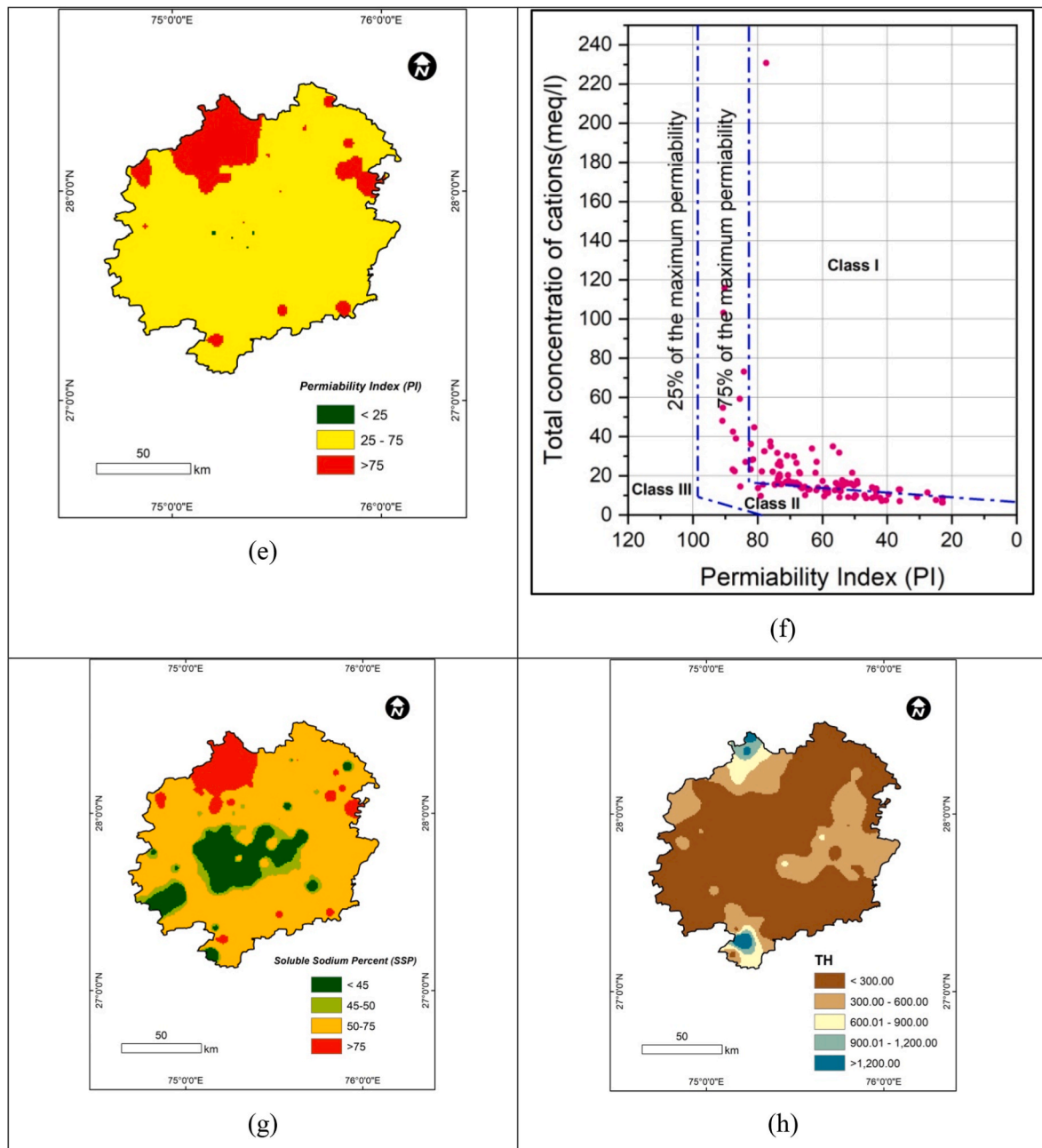


Fig. 6. (continued).

and halite dissolution and can be expressed using following expressions (10 and 11)

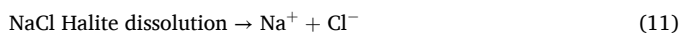
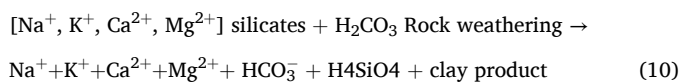


Fig. 4a indicated the trends of Na^+ versus Cl^- . Majority of samples (60%) were below the uniline which indicates reverse ion exchange. As the Cl^- concentration dominated over Na^+ , it can be inferred that as the source of Cl^- in the earth crust is non-geogenic, the excess Cl^- might also be non-geogenic in nature (Sahu et al., 2021). Majority of samples (60%) were below the uniline which indicates reverse ion exchange. Fig. 4b showed the trends of $\text{Ca}^{2+} + \text{Mg}^{2+}$ versus HCO_3^- and generally used to depict the dominant process between rock weathering or cation exchange as the groundwater quality controller. All the results above the

uniline were related to rock weathering or cation exchange (Rao et al., 2022). In this segment, the dominance of $\text{Ca}^{2+} + \text{Mg}^{2+}$ was more than the HCO_3^- . On the other hand all the results obtained below the uniline showed dominance of HCO_3^- over $\text{Ca}^{2+} + \text{Mg}^{2+}$ resulted in to release of HCO_3^- with close relation to feldspar minerals with H_2CO_3 in the groundwater. 85.83% of the total samples were above the 1:1 uniline showing the dominance of $\text{Ca}^{2+} + \text{Mg}^{2+}$ over HCO_3^- indicating rock weathering or cation exchange while only 6.6% of the samples were below the uniline. The result can also be justified by evaluating the trends of HCO_3^- versus $(\text{Cl}^- + \text{SO}_4^{2-})$ (Fig. 4c). Fig. 4d revealed the trends of $(\text{Ca}^{2+} + \text{Mg}^{2+})$ versus $(\text{HCO}_3^- + \text{SO}_4^{2-})$ where considerable number of samples were above the uniline indicating dominance of $\text{Ca}^{2+} + \text{Mg}^{2+}$. The result depicted the fact that excess $(\text{Ca}^{2+} + \text{Mg}^{2+})$ over HCO_3^- is the indication of reverse ion exchange due to origin of $(\text{Ca}^{2+} + \text{Mg}^{2+})$ from aquifer material (Athamena et al., 2022). From Fig. 4e it can be observed that significant number of samples were above the uniline of depicting

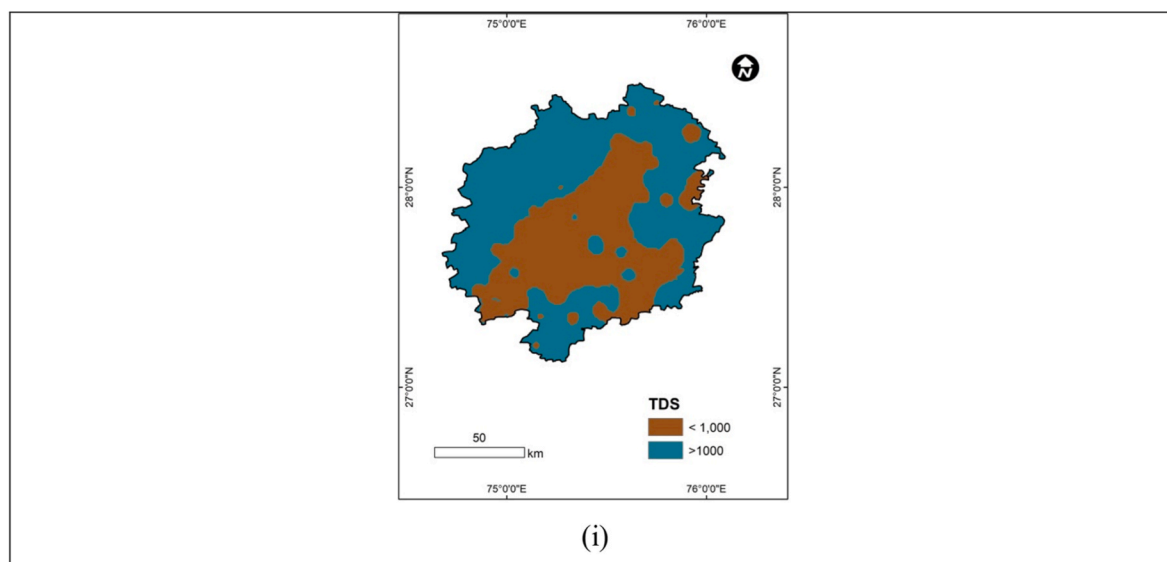


Fig. 6. (continued).

the dominance of $(Ca^{2+} + Mg^{2+})$ over $(Na^{+} + K^{+})$. The trends of $(Ca^{2+} + Mg^{2+})$ versus total cations, clearly showed the dominance of $(Ca^{2+} + Mg^{2+})$ over total cations as majority of samples were above the uniline (Fig. 4f). The obtained results most likely a consequence from increasing rock-water interactions (rock domination), weathering of host rocks, and chemical weathering of rock-forming minerals, which promote mineral breakdown and intensification of fluoride concentration in groundwater (Adimalla and Li, 2019).

3.5. Impact of ion exchange process on overall groundwater chemistry

Chloro-alkaline indices (CA-I) and (CA-II), developed by Schoeller (1967) are commonly used for identification of impact of ion exchange process on overall chemistry of groundwater. The expression of CA-1 and CA-2 are as follows (12 and 13).

$$\text{Chloro - alkaline index I (CAI - I)} = \frac{Cl^{-} - Na^{+} + K^{+}}{Cl^{-}} \quad (12)$$

$$\text{Chloro - alkaline index II (CAI - II)} = \frac{Cl^{-} - Na^{+} + K^{+}}{SO_4^{2-} + HCO_3^{-} + CO_3^{2-} + NO_3^{-}} \quad (13)$$

The obtained results were depicted in Table 6. 92.5% of the samples were categorized in the positive indices of CAI-I which confirms the reverse ion exchange process as the major controlling mechanism of the groundwater system (Fig. 4g) and possibly resulted into exchange between Na^{+} and K^{+} in the groundwater with Ca^{2+} and Mg^{2+} in the aquifer material (Schoeller, 1965, 1967).

3.6. Saturation index (SI)

Saturation Index (SI) is one of the most widely used tool for determining the stability of minerals between solid mineral state and aqueous phase of groundwater indicating the propensity towards precipitation or dissolution (Tomaszewska et al., 2018). The SI with respect to Halite, Gypsum and Calcite were computed using the following formula (14):

$$SI = \log \left(\frac{IAP}{K_{eq}} \right) \quad (14)$$

Here IAP is the product of the effective concentration of dissolved species.

K_{eq} is the equilibrium constant.

The maximum, minimum and average value of Halite (NaCl), Gypsum ($CaSO_4 \cdot 2H_2O$) and Calcite ($CaCO_3$) are represented in Table 7. From the result it can be observed that the minimum, maximum and average value of Halite (NaCl) was negative which indicated under saturation condition. Similar pattern was also reported by Zhang et al., (2020). On the other hand, both Calcite ($CaCO_3$) and Gypsum ($CaSO_4 \cdot 2H_2O$) depicted positive values indicating over saturation condition. This positive SI value of Calcite ($CaCO_3$) has a strong tendency of leaching F^{-} in the groundwater (Rusiniak et al., 2021; Subba Rao et al., 2020).

3.7. Water quality index (WQI)

The result obtained from the WQI analysis was divided into five categories viz. Excellent water Quality (<25.00), Good water quality (25.00–50.00), Poor water quality (50.01–75.00), Very poor water quality (75.01–100.00) and water unsuitable for drinking (>100.00). The map (Fig. 5a) showed the eastern and central portion of the study area of the groundwater quality was of good to excellent water quality where the western and southern portion, it was under poor water quality in the northern, western and some patches in southern portion.

3.8. Irrigation water quality index (IWQI)

Descriptive statistics of IWQI are tableted in Table 8. IWQI based on Sodium Absorption Ratio (SAR) (Formula No. 15, Table 2) can generally be divided into four categories: "excellent" (SAR <10 meq/l), "good" (SAR between 10 meq/l and 18 meq/l), "doubtful" (SAR between 18 meq/l and 26 meq/l), and "unsuitable" (SAR >26 meq/l). 68.33% and 18.33% of the total samples were categorized as 'excellent' and 'good' respectively while 9.17% and 4.17% of the total samples were associated with doubtful and unsuitable class (Fig. 5b and c). The IWQI map showed that Good and excellent water quality of groundwater was found in central and eastern portion while the peripheral region of the study area was under in poor water quality. Very poor water quality and unsuitable for irrigation category were found in northern portion and some patches in eastern and southern segments. (Fig. 5d). From the spatial point of view, central and eastern segment depicted excellent category of SAR. North, west and some smaller segments of south showed good category. In the north east and north west, doubtful as well as unsuitable category of SAR was observed (Fig. 6a). The Magnesium Absorption Ratio (MAR) was calculated using formula no.16 (Table 2) (Paliwal

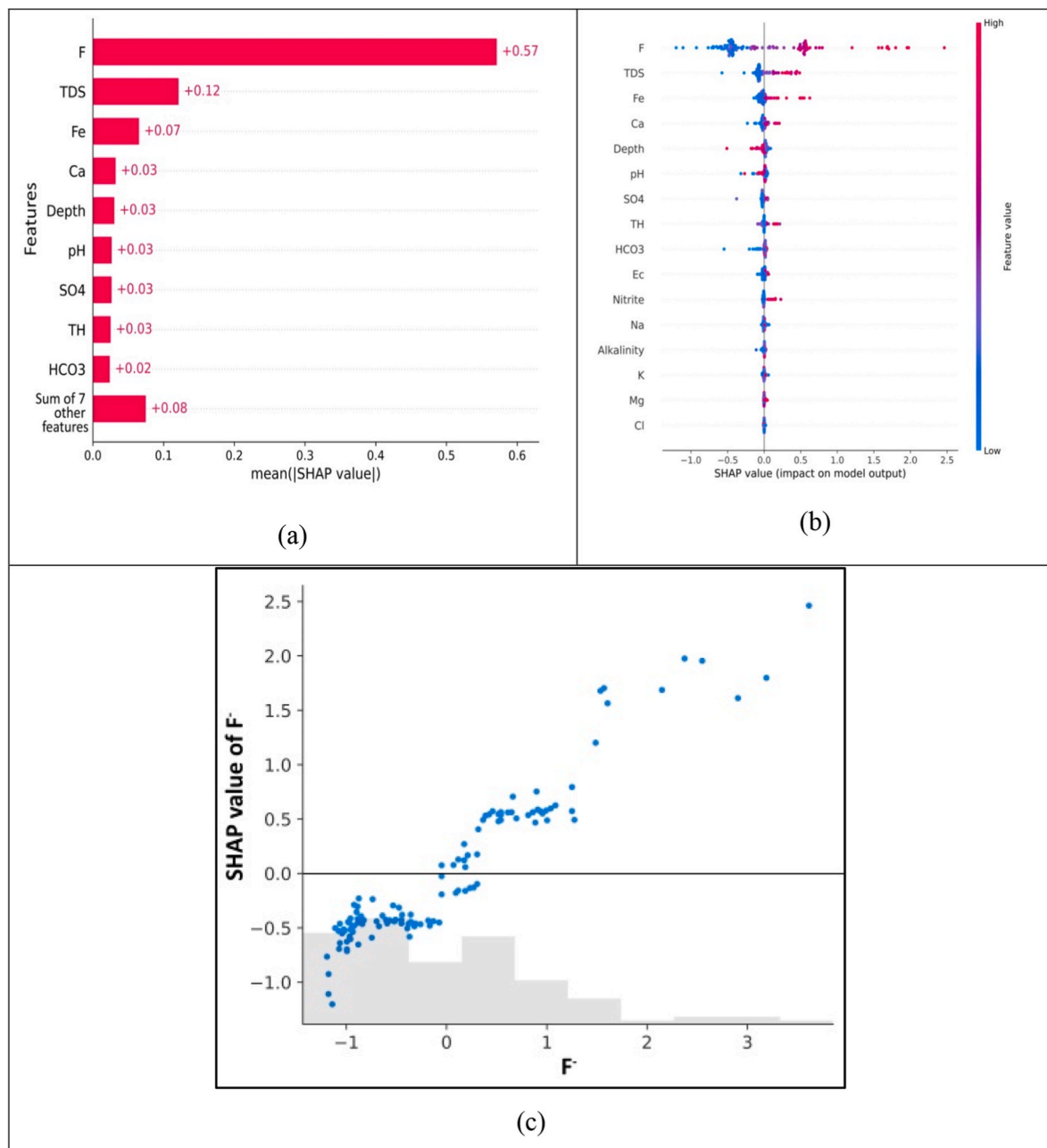


Fig. 7. (a) The order of importance of features of groundwater quality for drinking purpose, from most important to least important. (b) SHAP summary plot of feature variables of groundwater quality for drinking purpose. (c) Scatter plot showing significant feature values and SHAP values of F^- .

1972). If the MAR in water is more than 50, it is harmful for irrigation, and if it is less than 50, it is safe for irrigation (Houatmia et al., 2016). 54.17 % of the total area was under suitable types while 45.83 % of the region was associated with unsuitable for irrigation purposes. From the spatial point of view, northern and western segment was associated in suitable category while the eastern and central part was in unsuitable category (Fig. 6b). According to Residual Sodium Carbonate (RSC) categorization (Formula No. 17, Table 2), most of the segment of the study area was in suitable category of groundwater quality. Few patches in the north eastern and north western part were associated with marginally suitable to not suitable category (Fig. 6c). Formula no. 5 is used to compute the Kelly's Ratio (KR) (Kelly 1954). It is acceptable for irrigation if KR is < 1 , while it is not suitable for irrigation if KR is > 1 (Kelly, 1963) (Formula No. 18, Table 2). As per the classification of KR, 30% and 70% of the study area was in the suitable and unsuitable water quality types for irrigation uses respectively. The suitable groundwater

quality zone for irrigation purposes was mostly observed in the central and southern portion while rest of the part was associated with unsuitable category (Fig. 6d). As per the Permeability Index (PI) (Formula No. 19, Table 2), 23.33% of the total samples were classified under Class I (Suitable category) while 72.50% and 4.17% of the samples were categorized as Class II (Good category) and Class III or (Unsuitable category) respectively (Fig. 6e and f). Hence it can be observed that majority of the samples were associated with suitable to good category of PI. In the central patch of the study area showed the lower level of Soluble Sodium Percent (SSP) on the other hand in the northern periphery the level of SSP was relatively higher (Fig. 6g) (Formula No. 20, Table 2). Higher concentration of Total Hardness as CaCO_3 (TH) (Formula No. 21, Table 2) was observed in northern and southern portion while the central and eastern portion of the study area had low concentration (Fig. 6h). Except central and southern portion, the level of TDS was relatively higher throughout the region (Fig. 6i).

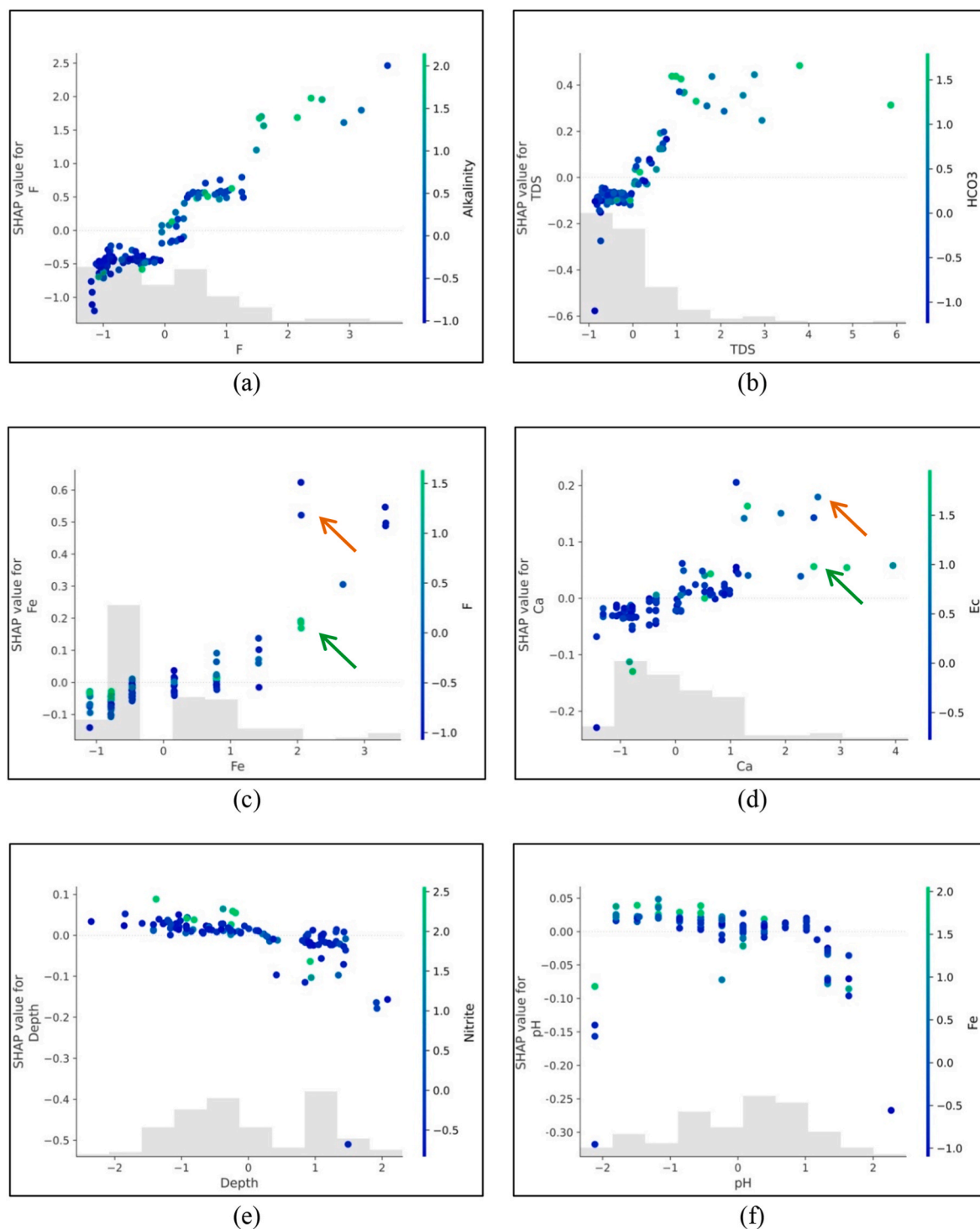


Fig. 8. SHAP dependence contribution plots for groundwater quality for drinking purpose (a) Effect of F^- on predicted water quality with respect to the Alkalinity (b) effect of TDS on predicted water quality with respect to the HCO_3^- (c) Effect of Fe^{2+} on predicted water quality with respect to the Fe^{3+} (d) Effect of Ca on predicted water quality with respect to the EC. (e) Effect of Depth on predicted water quality with respect to the Nitrite. (f) Effect of pH on predicted water quality with respect to the Fe.

3.9. Global interpretability of WQI with XAI

The variable importance plot ranks the variables by importance and mean SHAP value. F^- had the highest mean SHAP (+0.57), followed by TDS (+0.12) and Fe (+0.07). Rest other variables had lower SHAP values. Thus, F^- , TDS, and Fe contributed significantly to model prediction, while the other variables had relatively little impact (Fig. 7a). SHAP values presented by WQI variables were shown in Fig. 7b. Each

dot represented a single observation, and the colour denoted feature values: red for high, blue for low. Horizontal locations inferred that value connected with higher or lower prediction. F^- showed a wide range of positive and negative SHAP values. Higher F^- values with positive SHAP values (point observations extended to the right displayed red colour) and extended up to 2.5 mean SHAP value. On the other hand, low F^- values were mostly extended towards the negative SHAP value (up-to -1.0) (Point observation extended towards left hand side showed

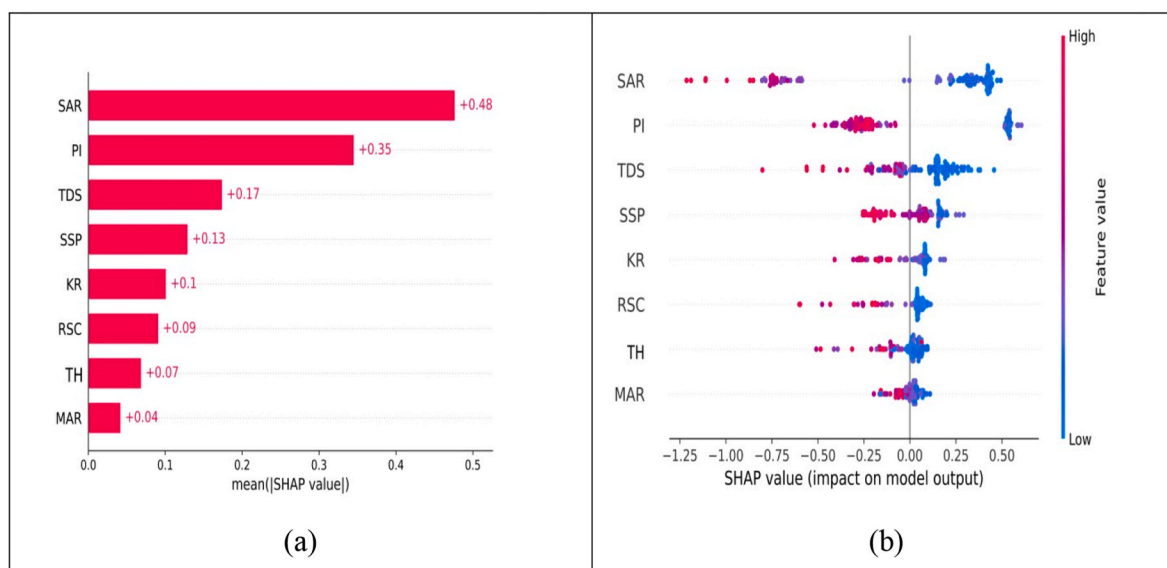


Fig. 9. (a) The order of importance of features of groundwater quality for irrigation purpose, from most important to least important. (b) SHAP summary plot of feature variables of groundwater quality for irrigation purpose.

higher intensity of blue colour). Hence it can be inferred that higher concentration of F^- value has higher predictability of overall water quality. In the similar way, TDS, Fe, Ca, TH and Nitrite had significant high positive influence on groundwater quality. On the other hand, Depth, pH, HCO_3^- was associated with significantly low negative SHAP values that showed a lower level of predictability with groundwater quality. Significant feature value and SHAP values of F^- depicted in Fig. Fig. 7c.

3.10. SHAP dependence contribution plots for WQI

To depict the pattern of effect as well as to understand if the effect of having a certain value relatively consistent, or does it varies significantly depending on the values of other features, SHAP dependence contribution plots are considered to be one of the important tools. We have considered the six most important variables. Fig. 8a to Fig. 8f depicted SHAP dependence contribution of significant WQI indices and strongly interacting features. In Fig. 8a–f dependence plot for the features, the points were coloured by values as represented in the colour bar on the right side respectively. The result showed that there was a strong impact of other features on the prediction, rather a single feature. Fig. 8a showed the effect of the F^- on predicted water quality with respect to the Alkalinity. The colour gradient of each dot represents the original value of the alkalinity from low (blue) to high (green). The plot showed that F^- and the target variable had roughly linear and positive relationship along with frequent interaction with Alkalinity. Fig. 8b also depicted similar pattern and inferred that TDS and HCO_3^- interacted frequently and showed a positive association with the water quality. In Fig. 8c two arrows indicated that the level of feature value of Fe was almost identical in both the observations but they had different impact on the entire model as respective SHAP values were different. That value caused one prediction to increase (orange arrow), and it caused the other prediction to decrease (green arrow). Similar kind of result can be observed in Fig. 8d also. Fig. 8e and f depicted non-linearity in the initial segment of the graph followed by a negative association.

3.11. Global interpretability of IWQI

Fig. 9a showed SAR as the most significant feature that significantly contributed (mean SHAP value of +0.48) in the groundwater quality prediction for irrigation purpose followed by PI (mean SHAP value =

+0.35), TDS (mean SHAP value = +0.17), SSP (mean SHAP value = +0.13), KR (mean SHAP value = +0.10), RSC (mean SHAP value = +0.09), TH (mean SHAP value = +0.07) and MAR (mean SHAP value = +0.04). Fig. 9b depicted the SHAP values involved in prediction of water quality for irrigation. From the model it can be deduced that SAR had a high negative influence on groundwater quality for irrigational purpose. In PI, majority of the positive SHAP values were concentrated near +0.50. Similar kind of pattern was observed in rest of the indices also and depicted negative SHAP value and negative correlation with the groundwater quality for irrigation.

3.12. SHAP dependence contribution plots for IWQI

The spread of the graph signified the influence of other features on the model rather than feature itself. Fig. 10a to Fig. 10f showed the dependence contribution plots for IWQI. In all the plots except Fig. 10f, it can be depicted that in the initial instances, higher SHAP values were associated with relatively lower level of strongly interacting indices. For an instance, in Fig. 10a, the SHAP value of SAR and KR as strongly interacting feature is portrayed. It can be clearly depicted that higher SHAP values of SAR (above 0) were associated with lower level of KR. On the other hand, the lower SHAP value of SAR (below 0) were associated with higher level of KR. In Figs. 10d and f the orange and green arrows showed that even with similar level of SSP, it can have different effect on the entire model as the SHAP values are different.

4. Conclusion

The present study aimed to evaluate the groundwater hydro-geochemistry in the arid and semi-arid region of Rajasthan, India. It also focused on water quality appraisal for drinking and irrigation purpose combined with SHAP value based eXplainable Artificial Intelligence. The results indicated that the hydro-geochemical facies of the groundwater were mostly characterised by a Na–Cl composition and suggested a prevalence of evaporation processes. The research indicated that a high Saturation Index of Calcite ($CaCO_3$) could be closely linked to the leaching of F^- in the groundwater. Groundwater Quality Index for drinking purpose illustrated that central segment of the study area was associated with better water quality than the peripheral region. Feature extraction by XGBoost algorithm showed impact on the overall model prediction with highest SHAP value in the case of F^- . SHAP dependence

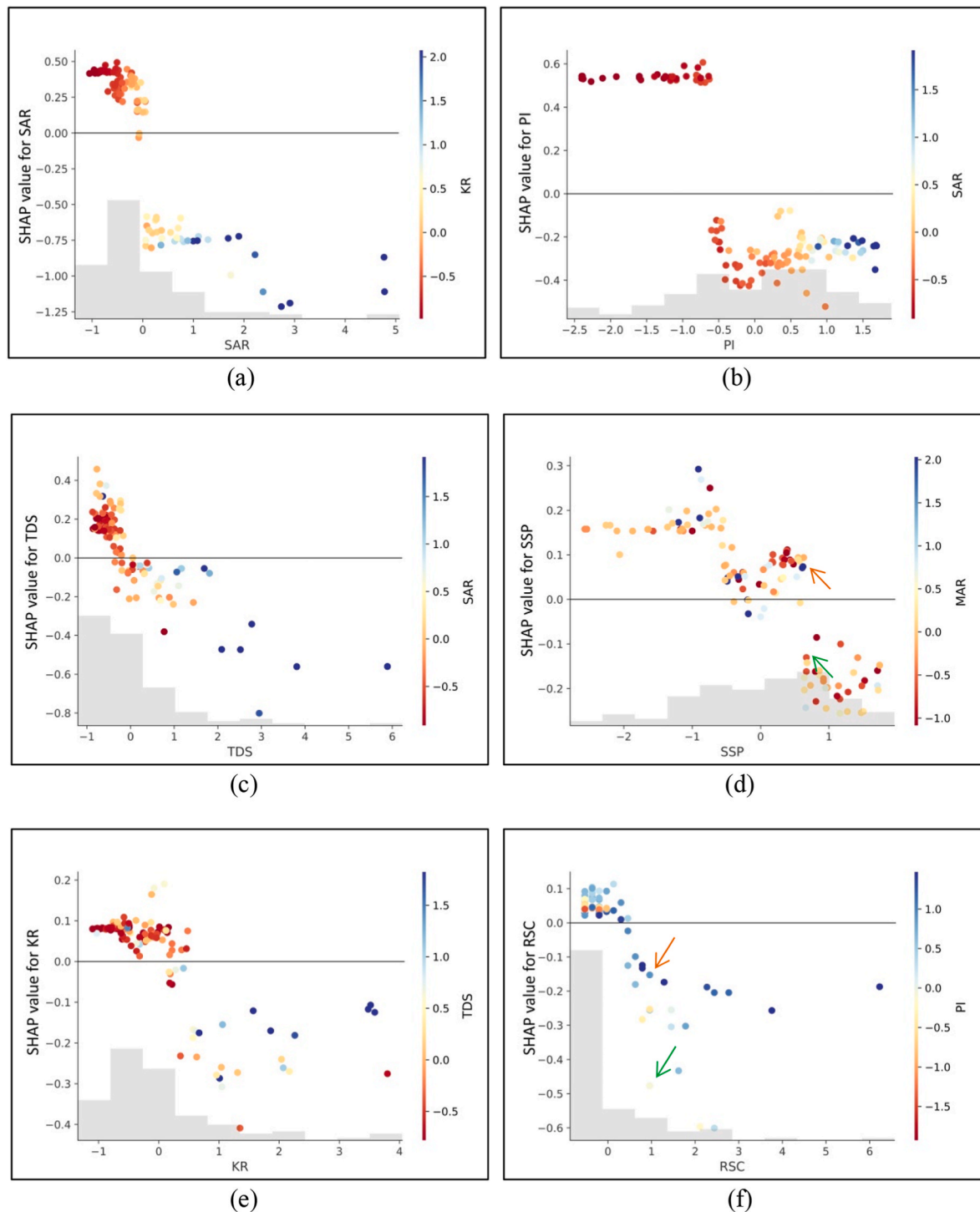


Fig. 10. SHAP dependence contribution plots for groundwater quality for irrigation purpose (a) Effect of SAR on predicted water quality with respect to the KR (b) effect of PI on predicted water quality with respect to the SAR (c) Effect of TDS on predicted water quality with respect to the SAR (d) Effect of SSP on predicted water quality with respect to the MAR. (e) Effect of KR on predicted water quality with respect to the TDS. (f) Effect of RSC on predicted water quality with respect to the PI.

contribution plots, in the majority of instances, the relationship exhibited a non-linear pattern characterised by a wide vertical span, indicating the influence of synergistic effects from parameters in predicting the model. Among multiple indices, SAR and PI showed higher average SAHP value in terms of groundwater quality for irrigation purpose. According to our knowledge, this is the first attempt for depicting groundwater quality combining SHAP based XGBoost algorithm in arid and semi-arid segment of Rajasthan, India. The finding of the study will definitely enhance the understanding of groundwater

behaviour not only in the present study area but also at global scale with similar climatic condition. The implications will also be helpful for the policy makers for future management of groundwater for drinking as well as for irrigation purpose. In the future, the study would be further extended by evaluating the performance of the model presented in this study compared to other machine learning models in the temporal dimension to comprehend groundwater hydro-geochemistry dynamics in the arid and semi-arid environment.

CRediT authorship contribution statement

Sunita: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Tathagata Ghosh:** Writing – original draft, Visualization, Supervision, Software, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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